Nikhil Kolhe

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Education

Dr. D.Y. Patil Institute Of Technology, Pune. June 2024
B.E.(A.I. And Data Science); CGPA:8.5 *Pune, Maharashtra*

Skills

Programming Languages: Python, SQL

Generative AI & Agentic AI: LLMs, LangChain, LangGraph, MCP, FastMCP, Multi-Agent Systems, RAG, Embeddings, Vector Search, Prompt Engineering, Tool Calling, Structured Outputs, Human-in-the-Loop Workflows

LLM Providers & APIs: OpenAI, Google Gemini, Groq, REST APIs, FastAPI, Flask, GraphQL

Databases & Deployment: SQLite, MySQL, PostgreSQL, Snowflake, SQLAlchemy, Chroma, FAISS, Git/GitHub

Machine Learning, Data & BI: scikit-learn, Pandas, NumPy, Deep Learning (RNN/LSTM), NLP, OCR, Databricks, Power BI, Tableau, TIBCO Spotfire

Experience

I2E Consulting 2024 – Present
Associate Developer *Remote*

Agentic AI Resume Analyzer | *Python, Gemini, MCP, LangGraph, FAISS, FastAPI, Streamlit, Tesseract OCR, Docker*

- Built an end-to-end agentic AI platform that ingests resumes (PDF, DOCX, scanned images via OCR), extracts schema-validated data, and scores fit against job descriptions using embeddings and FAISS similarity; developed custom **MCP servers** exposing parser, ATS-scoring, and skill-gap tools that a LangGraph workflow (Parser, Evaluator, Rewriter) chains to generate targeted resume improvements, served via FastAPI and Streamlit and containerized with Docker.

Clinical Study Resource Demand Forecasting (Pfizer) | *Databricks, PySpark, Delta Lake, MLflow, XGBoost/LightGBM*

- Built and deployed a production regression model on Databricks to predict study-wise monthly resource demand for Pfizer clinical studies, enabling proactive resource planning across the study portfolio. Engineered features on Delta Lake from study attributes, timelines, and historical resource allocation, tracked in MLflow, and automated monthly inference and retraining with Databricks Workflows. improved forecast accuracy by 29% over the baseline.

Time Card Report | *Power BI, Power Automate, Data Gateway*

- Developed interactive Power BI reports to analyze employee working hours, overtime, project allocation, resource utilization, and time-reporting compliance. Created an executive infographic report presenting key workforce insights for management and supporting data-driven decisions.

Projects

Drug Discovery Agent | *Python, LangGraph, MCP, FastAPI, OpenAI, Gemini, RAG*

- Built a multi-agent drug-target prioritization system that integrates biomedical evidence from DepMap, Open Targets, Pharos, and Europe PMC using MCP-connected data tools. Implemented a stateful LangGraph workflow for gene validation, parallel evidence collection, data normalization, evidence scoring, and cross-source conflict detection. Developed evidence-grounded research report generation and RAG-based follow-up queries, with source-traceable results, human-in-the-loop review, and FastAPI endpoints.

Clinical Study Assistant Agent | *Python, LangGraph, FastMCP, FastAPI, scikit-learn*

- Integrated trained KPI prediction models as MCP tools to predict clinical study KPIs, retrieve similar studies, and compare study results. Developed a LangGraph agent that interprets natural-language queries, selects appropriate tools, and returns results through REST APIs.

AI-Powered Blog Generator (LLM) | *Llama 3.1 8B, AWS Bedrock, RAG, Streamlit*

- Developed an AI-powered content generator leveraging Llama 3.1 8B, AWS Bedrock, and RAG for real-time fine-tuning on dynamic datasets. Integrated with Streamlit for a seamless user interface, the system automated blog creation, editing, and publishing, increasing efficiency in content marketing and engagement for enterprises.

Certifications & Publications

AWS Certified AI Practitioner (AIF-C01)
IEEE Publication: *Fun Soundify: Music Generation powered by AI*